

Will Varner

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Education

University of Georgia

Athens, GA

Bachelor of Science in Computer Science; GPA: 3.8

Expected May 2027

Honors: UGA Presidential Scholar, Zell Miller Scholar, Order of Omega (Top 5%)

Experience

Roblox

San Mateo, CA

Software Engineering Intern

May 2026 – August 2026

- Built an LLM-driven semantic IR front end in Python, replacing 9,700 lines of rigid parsers with declarative behavior catalogs adopted across 40+ production repositories in 8 engineering orgs.
- Resolved platform reliability bugs in BuilderAI: diagnosed a TTLCache credential desync causing 401s during token renewal and eliminated false-positive reporting on reverted testgen runs.
- Surfaced 7 owner-confirmed production defects via behavior reconciliation, directly driving 3 merged production patches and the deletion of ~2,000 lines of deprecated routing logic.
- Integrated behavior contracts into BuilderAI's LangGraph/C# testgen pipeline, lifting critical behavioral obligation coverage by +36.4% in a controlled same-commit, identical-budget A/B benchmark.
- Engineered a differential evaluation harness using order-reversed mutation fixtures to eliminate judge bias, validating spec drift and filtering out framework-level bloat across 5 LLM model tiers.

Mobility Systems Control Laboratory

Athens, GA

Undergraduate AV Research Lead

January 2025 – Present

- Built a closed-loop red-teaming pipeline coupling a 20Hz C++ telemetry daemon with CARLA 0.9.15 and SUMO co-simulation via POSIX shared-memory IPC.
- Integrated NVIDIA Alpamayo 2 Super (34B VLA), ingesting multi-camera streams to record trajectory traces and natural-language reasoning tokens at 50ms deterministic tick rates.
- Engineered an adversarial scenario mutator in Python that dynamically perturbs pedestrian crossing trajectories and occlusion angles conditioned on real-time ego vehicle kinematics.
- Benchmarked failure yield rates and minimum time-to-collision across 40+ dynamic collision topologies, sustaining deterministic 50ms evaluation loops across continuous co-simulation runs.
- Identified safety-critical failure modes in a 34B VLA model where the network accurately verbalized imminent collision hazards yet simultaneously issued contradictory steering commands.

Leidos

Atlanta, GA

Software Engineering Intern

May 2025 – January 2026

- Ported 15+ backend nodes from CARMA-CARLA Bridge to ROS 2 (rclpy), eliminating legacy bridge middleware to achieve direct, sub-5ms communication with the CARMA Platform.
- Engineered carla_ros2_bridge in C++ and Python, authoring real-time sensor and vehicle interfaces for CARLA 0.10.0 to enable direct, high-throughput ROS 2 co-simulation.
- Prototyped a unified XML-RPC protocol replacing TraCI for actor and sensor synchronization, reducing serialization overhead by 50% across interaction pipelines.

Technical Skills

- **Languages:** C++, C, Python, Go, C#, SQL, Lua, TypeScript
- **Simulation & Robotics:** ROS 2 (rclpy), CARLA (LibCarla), SUMO, POSIX Shared Memory IPC
- **AI & Backend Infrastructure:** LangGraph, Pydantic, gRPC, .NET, OpenCV
- **Systems & Observability:** Linux/Unix, Docker, VictoriaMetrics, Kibana, CI/CD (GitHub Actions), AWS, GCP